

USAA Fraud Detection – Project Timeline & Next Steps

1) Finalize Supabase Data Pipeline

Verify data completeness and embeddings.

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| • Confirm 1,717 rows in press_releases_clean |
| • Spot-check that embedding column is populated (not NULL) |
| • Fix any missing content/metadata and re-embed if needed |

2) Streamlit Semantic Search App (MVP)

Allow users to search semantically and view top matches.

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| • Text input → Supabase RPC (match_press_releases) |
| • List top N matches: title, date, similarity, link |
| • Optional filters: year and keyword |

3) Analytics & Visualization (Jupyter)

Build supporting visuals for stakeholders.

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| • Year-over-year counts of fraud/AML articles |
| • Keyword/phrase frequencies |
| • Topic clusters from embeddings (UMAP/KMeans) |

4) Presentation Prep

Communicate results effectively to USAA & class.

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| • Pipeline diagram (scrape → clean → embed → search) |
| • Demo the Streamlit app with AML/fraud queries |
| • Include charts/trends & key insights |

5) Optional: RAG Summaries

Summarize retrieved articles with an open-source LLM.

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| • Keep short summaries (2–3 sentences) |
| • Cite source URLs under each summary |
| • Add to Streamlit as a toggle later |

Suggested Order & Timing

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| 1. Week 1: Verify data + embeddings (Phase 1) |
| 2. Week 1–2: Streamlit MVP (Phase 2) |
| 3. Week 2: Jupyter visuals (Phase 3) |
| 4. Week 3: Polish presentation + dry runs (Phase 4) |
| 5. If time permits: RAG summaries (Phase 5) |